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Coursework: PSD lab (ECDII – EEEN20222)
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1. Aims

The purpose of this experiment was to investigate the operating principles of phase-sensitive demodulation (PSD) and low-pass filtering through practical circuit construction and measurement. Specifically, the experiment aimed to:

- Exercise use of bench instruments, including a DC power supply, function generator, digital oscilloscope and a digital multimeter (DMM)
- Design, build and test circuits on a breadboard: a phase-shifter, an inverter, a switch-based PSD and a first-order low-pass filter (LPF)
- Characterise the amplitude-frequency response of the LPF, compare the measured Bode plot with the theoretical predictions and quantify the sources of error
- Understand how a PSD extracts a DC signal whose magnitude and polarity depend on the amplitude and relative phase of two input signals at the same frequency

2. Circuit schematics

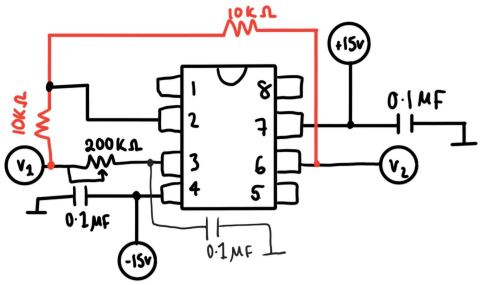


Figure 2.1: Phase Shifter Schematic Diagram

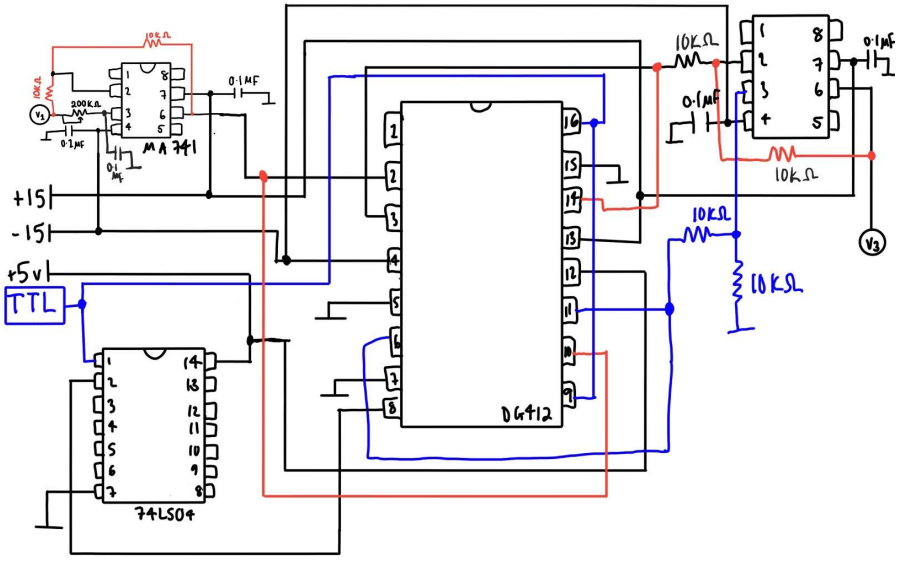


Figure 2.2: Switch-based PSD Schematic

3. Experimental results and analysis

3.1 Signal generator output (step 1)

The function generator was configured to produce a 1kHz sine wave at 2Vp-p with DC offset = 0 from its output terminal and a synchronous square wave from its sync terminal. Both waves and their amplitudes were examined on the oscilloscope simultaneously, as shown in figure 3.1.

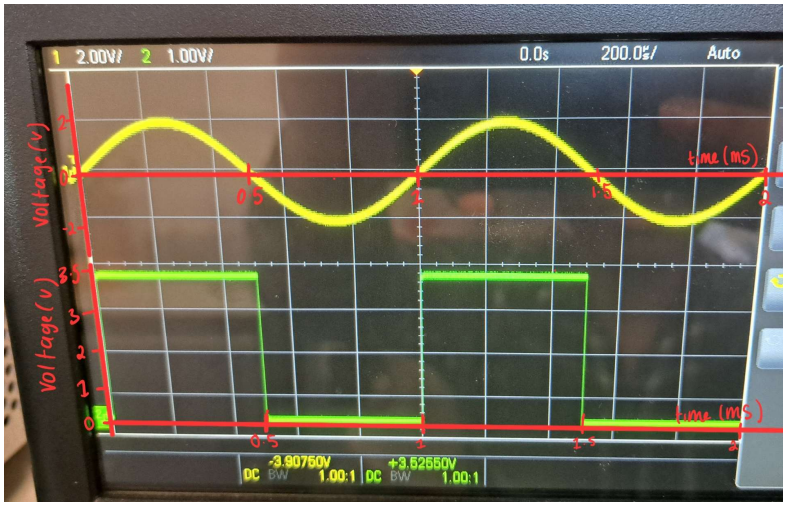


Figure 3.1: Oscilloscope Screenshot of Sine and Square wave from function generator

3.2 Phase shifter (step 2-3)

The phase shift circuit in figure 2.1 was built using a µA741 op-amp with two 10kΩ resistors, three 0.1µF capacitors and a 0-200kohm trimmer resistor
Adjusting the trimmer from 0kohms (figure 3.2.1) to its max resistance (figure 3.2.2), the range of the phase shift was observed to be nearly 180 lagging relative to the input, confirming the unity-gain operation across the full range

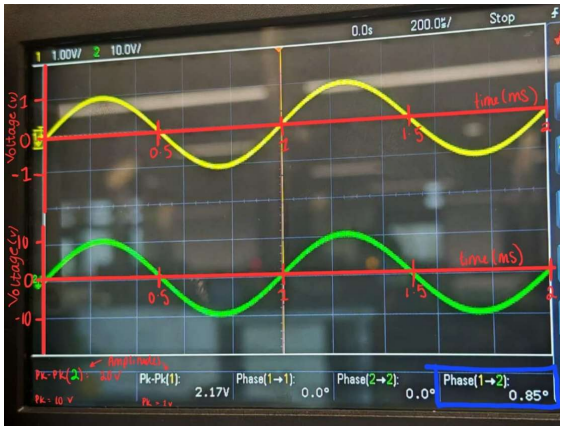


Figure 3.2.1: Minimum Phase Shift

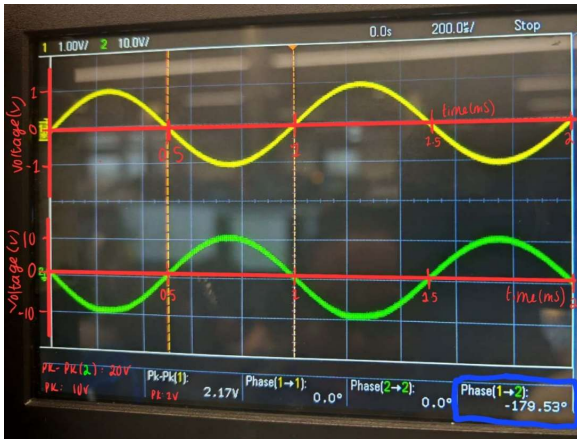


Figure 3.2.1: Maximum Phase Shift

3.3 Switch-based PSD Output at 0° phase difference (step 6)

The PSD circuit was constructed using the phase shifter, a DG412 quad switch, a second µA741 op-amp (and relevant resistors and capacitors) and a 74LS04 inverter. The PSD output resembled a full-wave rectified waveform with peaks pointing upwards, as shown in Figure 3.3

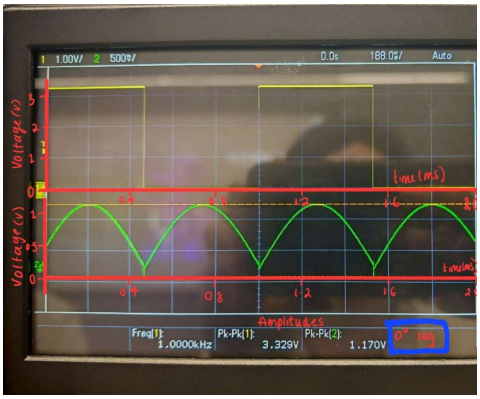


Figure3.3: Switch-Based PSD at 0°

3.4 Switch-based PSD Output at 90° phase difference (step 7)

The phase shifter is adjusted using the trimmer, so the phase difference between the sine wave and the square wave is now 90°. The output wave is shown in figure 3.4 with repetitive half-waves



Figure 4: Switch-Based PSD at 90°

3.5 Low-Pass Filter (step 8)

The low-pass filter in Figure 4 of LabScript was built using a third µA741 op amp, two 160kΩ resistors and a 0.1 µF feedback capacitor. The oscilloscope in Figures 3.5.2 and 3.5.3 captures the frequency response at 10Hz and 1kHz, respectively.

Its theoretical cut-off frequency is:

$$F_o = \frac{1}{2\omega CR} = \frac{1}{2 \times \pi \times 160000 \times (0.1 \times 10^{-6})} = 9.95Hz$$

Table 1 below presents the measured and theoretical values and the relative error.
The equation used to calculate the theoretical output:

$$V_o = V_i \times \frac{1}{\sqrt{1 + \frac{f^2}{{f_o}^2}}}$$

Table 1: Frequency response of LPF

Frequency (Hz)	Measured input (Vp-p)	Measured Output (Vp-p)	Theoretical output (Vp-p)	Measured value (dB)	Theoretical value (dB)	Relative error (%)
1	2.17	2.04	1.99	-0.537	-0.0437	1128
2	2.17	1.97	1.96	-0.840	-0.172	388
4	2.13	1.86	1.86	-1.18	-0.651	81.3
6	2.05	1.69	1.71	-1.68	-1.35	24.4
10	2.13	1.41	1.41	-3.58	-3.03	18.2
20	2.05	0.890	0.891	-7.25	-7.03	3.13
40	2.05	0.478	0.483	-12.65	-12.3	2.85
60	2.13	0.320	0.327	-16.46	-15.7	4.84
100	2.17	0.188	0.198	-21.2	-20.1	5.47
200	2.17	0.100	0.0993	-26.7	-26.08	2.38
400	2.13	0.0488	0.0497	-32.8	-32.09	2.21
600	2.17	0.0341	0.0332	-36.1	-35.6	1.40
1000	2.17	0.0219	0.0199	-39.9	-40.05	-0.375

By changing the frequency, the approximate -3dB frequency is 10 Hz and the amplitude is 1.42V which is the cut-off frequency

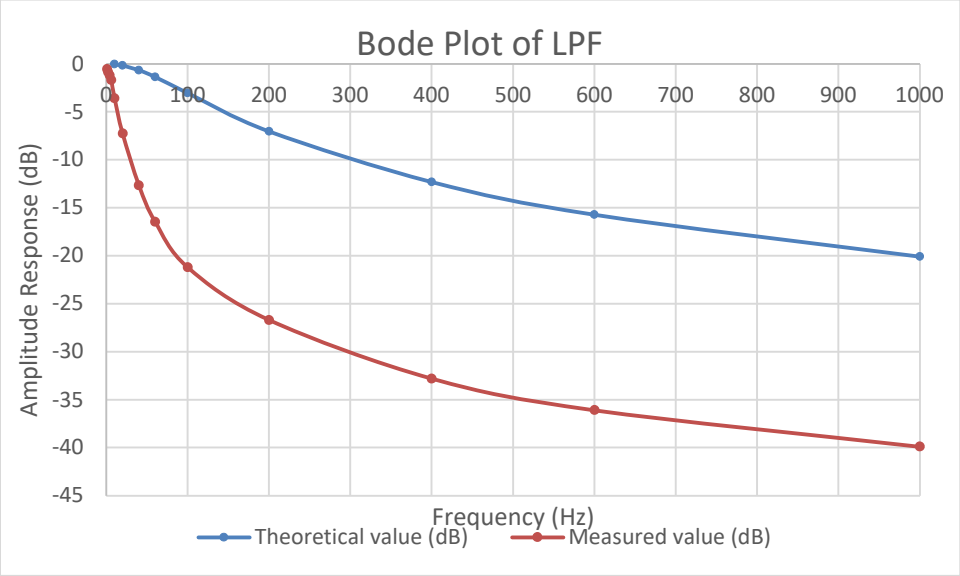


Figure 3.5.1: Bode Plot of LPF

In Figure 3.5.1, the theoretical amplitude response displays the zoomed in Bode plot for both the theoretical and measured amplitude responses and the cut-off can be observed to be -3 dB and a roll of beyond this point. The measured data points follow the theoretical curve closely at higher frequencies, but at lower frequencies, there are large discrepancies.

Sources of error could be due to contact resistance in the breadboard, oscilloscope reading uncertainty and component tolerances. Improvements could be made by using more precise components with lower tolerances and higher-performance op-amps.

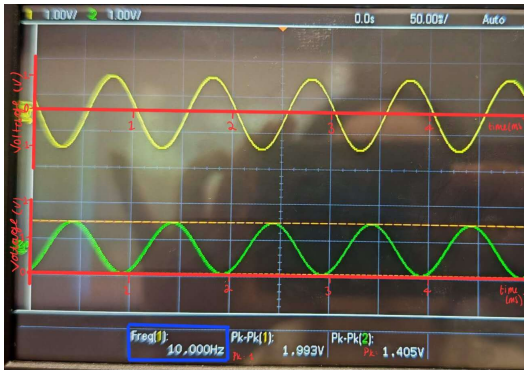


Figure3.5.2: LPF at 10Hz

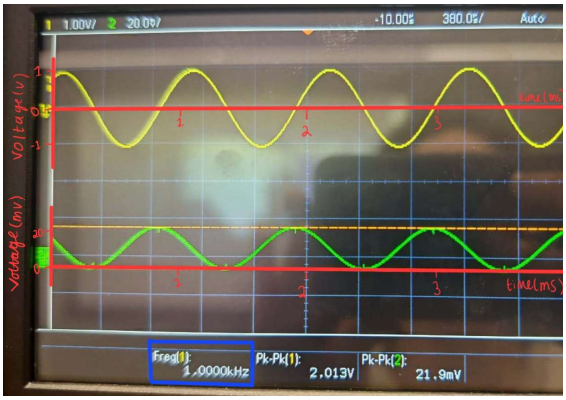


Figure 3.5.3: LPF at 1000Hz

3.5 DC Output (step 10)

With the LPF connected to the PSD output and a 1kHz frequency, the DMM measured the DC output voltage for two phase settings:

Table 2: DC value with different phases

Phase Difference	0°	90°
Measured DC value	±0.635	0.0231

4 Answers to questions

Question 1: If 90° phase delay is required for a 1 kHz sine wave, calculate the resistor value of the trimmer R in Fig. 2 and show the detailed calculation procedure. Note that for engineering, 3 significant digits are sufficiently accurate.

The equation for the phase shifter is:

$$\Phi = -2\tan^{-1}(\omega CR)$$

Phase delay of 90° and solving for R:

$$-90° = -2\tan^{-1}(\omega CR)$$

$$\tan^{-1}(\omega CR) = 45° \rightarrow \omega CR = \tan(45) = 1$$

$$R = \frac{1}{\omega C} = \frac{1}{2\pi \times 1000 \times (0.1 \times 10^{-6})} = 1591.5 \ldots = 1.59k\Omega$$

Question 2: In session 2, step 6, if you have seen 0° PSD signal, how can you obtain 180° PSD signal? Or if you have seen 180° PSD signal, how can you obtain 0° PSD signal?

A 0° PSD signal was seen. To obtain a 180° signal, the resistor trimmer needs to be adjusted to its maximum until the displayed waveform inverts

Question 3: With a cut-off frequency of 10 Hz of the LPF as shown in Fig. 4 (labscript), calculate how many times a 1 kHz sine input signal is attenuated after passing through the LPF circuit? If the cut-off frequency is increased to 100 Hz, how many times is the signal attenuated?

At F_c = 10Hz

$$Gain = \frac{1}{\sqrt{1 + \frac{1000^2}{10^2}}} \approx 0.01$$

The signal is attenuated by a factor of 0.01 (the output is 100 smaller than the input)

At F_c = 100Hz

$$Gain = \frac{1}{\sqrt{1 + \frac{1000^2}{100^2}}} \approx 0.1$$

The signal is attenuated by a factor of 0.1 (the output is 10 smaller than the input)

Question 4: What is the theoretical DC output of the PSD you built in the lab if the phase angle is 60°?

$$V_{DC} = \frac{2A}{\pi}\cos(\phi) = 0.38\text{V}$$

$$A = \frac{V_{p-p}}{2} = \frac{2}{2} = 1$$

5 Conclusions

This experiment successfully demonstrated the operating principle of a switch-based PSD followed by a first-order active low-pass filter. The phase-shifter circuit provided an adjustable phase delay from 0° to 180° by varying the trimmer resistance, confirming the theoretical relationship $\phi = -2\tan^{-1}(\omega CR)$. The switch-based PSD correctly implemented synchronous demodulation with a 0° phase difference, the output resembled a full-wave rectifier waveform, and at 90°, there were repeated half-waves. The LPF exhibited a first-order and the frequency response of the measured values was compared to the theoretical at lower frequencies, showing greater error. Overall, the experiment confirmed that a switch-based PSD followed by a low-pass filter can extract a DC voltage proportional to Cos(ϕ) where ϕ is the phase difference between the input sine wave and the reference square wave.